

Lab07

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```
path <- "\\smb/ra281972/Downloads/ME315"
fname <- file.path(path, "database.sqlite3")

conn <- dbConnect(RSQLite::SQLite(), fname)
```

```
SELECT
  i.name AS professor,
  SUM(gd.a_count * 4.0 +
      gd.ab_count * 3.5 +
      gd.b_count * 3.0 +
      gd.bc_count * 2.5 +
      gd.c_count * 2.0 +
      gd.d_count * 1.0 +
      gd.f_count * 0.0) * 1.0 /
  SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count +
      gd.d_count + gd.f_count) AS gpa_medio,
  SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count +
      gd.d_count + gd.f_count) AS total_alunos
FROM instructors i
INNER JOIN teachings t ON i.id = t.instructor_id
INNER JOIN sections sec ON t.section_uuid = sec.uuid
INNER JOIN course_offerings co ON sec.course_offering_uuid = co.uuid
INNER JOIN grade_distributions gd ON co.uuid = gd.course_offering_uuid
INNER JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
INNER JOIN subjects s ON sm.subject_code = s.code
WHERE s.abbreviation = 'STAT'
GROUP BY i.id, i.name
HAVING total_alunos > 0;
```

Table 1: Displaying records 1 - 10

professor	gpa_medio	total_alunos
JUN SHAO	3.662829	608
MICHAEL FERRIS	2.995000	400
JOSEP GINEBRA	3.313954	258
ANATOLE BECK	2.924657	73
ZHANG CHUNMING	3.563291	948
ISMOR FISCHER	3.077066	17472
PHILIP WOOD	2.982262	451
JONATHON PETERSON	2.776923	65
KYUNGMAN KIM	3.638889	198
NELLIE LAUGHLIN	3.164021	378

#professor mais difícil

```

SELECT
  i.name AS professor,
  SUM(gd.a_count * 4.0 +
    gd.ab_count * 3.5 +
    gd.b_count * 3.0 +
    gd.bc_count * 2.5 +
    gd.c_count * 2.0 +
    gd.d_count * 1.0 +
    gd.f_count * 0.0) * 1.0 /
  SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count +
    gd.d_count + gd.f_count) AS gpa_medio,
  SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count +
    gd.d_count + gd.f_count) AS total_alunos
FROM instructors i
INNER JOIN teachings t ON i.id = t.instructor_id
INNER JOIN sections sec ON t.section_uuid = sec.uuid
INNER JOIN course_offerings co ON sec.course_offering_uuid = co.uuid
INNER JOIN grade_distributions gd ON co.uuid = gd.course_offering_uuid
INNER JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
INNER JOIN subjects s ON sm.subject_code = s.code
WHERE s.abbreviation = 'STAT'
GROUP BY i.id, i.name
HAVING total_alunos > 0
ORDER BY gpa_medio ASC
LIMIT 1;

```

Table 2: 1 records

professor	gpa_medio	total_alunos
JAMES D KUELBS	2.607692	65

#professor mais facil

```
SELECT
  i.name AS professor,
  SUM(gd.a_count * 4.0 +
      gd.ab_count * 3.5 +
      gd.b_count * 3.0 +
      gd.bc_count * 2.5 +
      gd.c_count * 2.0 +
      gd.d_count * 1.0 +
      gd.f_count * 0.0) * 1.0 /
  SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count +
      gd.d_count + gd.f_count) AS gpa_medio,
  SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count +
      gd.d_count + gd.f_count) AS total_alunos
FROM instructors i
INNER JOIN teachings t ON i.id = t.instructor_id
INNER JOIN sections sec ON t.section_uuid = sec.uuid
INNER JOIN course_offerings co ON sec.course_offering_uuid = co.uuid
INNER JOIN grade_distributions gd ON co.uuid = gd.course_offering_uuid
INNER JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
INNER JOIN subjects s ON sm.subject_code = s.code
WHERE s.abbreviation = 'STAT'
GROUP BY i.id, i.name
HAVING total_alunos > 0
ORDER BY gpa_medio DESC
LIMIT 1;
```

Table 3: 1 records

professor	gpa_medio	total_alunos
SRIKANTHMADHAVAN ARAVAMUTHAN	4	16

```

SELECT
  c.name AS nome_disciplina,
  c.number AS numero_disciplina,
  SUM(gd.a_count * 4.0 +
      gd.ab_count * 3.5 +
      gd.b_count * 3.0 +
      gd.bc_count * 2.5 +
      gd.c_count * 2.0 +
      gd.d_count * 1.0 +
      gd.f_count * 0.0) * 1.0 /
  SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count +
      gd.d_count + gd.f_count) AS gpa_medio,
  SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count +
      gd.d_count + gd.f_count) AS total_alunos
FROM courses c
INNER JOIN course_offerings co ON c.uuid = co.course_uuid
INNER JOIN grade_distributions gd ON co.uuid = gd.course_offering_uuid
INNER JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
INNER JOIN subjects s ON sm.subject_code = s.code
WHERE s.abbreviation = 'STAT'
GROUP BY c.uuid, c.name, c.number
HAVING total_alunos > 0;

```

Table 4: Displaying records 1 - 10

nome_disciplina	numero_disciplina	gpa_medio	total_alunos
Special Topics in Statistics	692	3.872378	286
Theory of Probability II	734	3.562500	64
Applied Multivariate Analysis	456	3.437126	167
Applied Categorical Data Analysis	421	3.328947	304
Introduction to Theory and Methods of Mathematical Statistics I	311	3.146679	2710
Nonparametric Statistics and Machine Learning Methods	841	4.000000	24
Introductory Applied Statistics for the Life Sciences	371	3.215721	9109
Statistical Methods for Spatial Data	575	3.752336	107
Generalized Linear Models	851	3.680000	25
Introduction to Probability and Mathematical Statistics I	309	3.026515	1188

#disciplina mais dificil

```
SELECT
  c.name AS nome_disciplina,
  c.number AS numero_disciplina,
  SUM(gd.a_count * 4.0 +
      gd.ab_count * 3.5 +
      gd.b_count * 3.0 +
      gd.bc_count * 2.5 +
      gd.c_count * 2.0 +
      gd.d_count * 1.0 +
      gd.f_count * 0.0) * 1.0 /
  SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count +
      gd.d_count + gd.f_count) AS gpa_medio,
  SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count +
      gd.d_count + gd.f_count) AS total_alunos
FROM courses c
INNER JOIN course_offerings co ON c.uuid = co.course_uuid
INNER JOIN grade_distributions gd ON co.uuid = gd.course_offering_uuid
INNER JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
INNER JOIN subjects s ON sm.subject_code = s.code
WHERE s.abbreviation = 'STAT'
GROUP BY c.uuid, c.name, c.number
HAVING total_alunos > 0
ORDER BY gpa_medio ASC
LIMIT 1;
```

Table 5: 1 records

nome_disciplina	numero_disciplina	gpa_medio	total_alunos
Introduction to the Theory of Probability	431	2.886374	2231

#disciplina mais facil

```
SELECT
  c.name AS nome_disciplina,
  c.number AS numero_disciplina,
  SUM(gd.a_count * 4.0 +
      gd.ab_count * 3.5 +
      gd.b_count * 3.0 +
      gd.bc_count * 2.5 +
```

```

        gd.c_count * 2.0 +
        gd.d_count * 1.0 +
        gd.f_count * 0.0) * 1.0 /
SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count +
gd.d_count + gd.f_count) AS gpa_medio,
SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count +
gd.d_count + gd.f_count) AS total_alunos
FROM courses c
INNER JOIN course_offerings co ON c.uuid = co.course_uuid
INNER JOIN grade_distributions gd ON co.uuid = gd.course_offering_uuid
INNER JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
INNER JOIN subjects s ON sm.subject_code = s.code
WHERE s.abbreviation = 'STAT'
GROUP BY c.uuid, c.name, c.number
HAVING total_alunos > 0
ORDER BY gpa_medio DESC
LIMIT 1;

```

Table 6: 1 records

nome_disciplina	numero_disci- plina	gpa_medio	to- tal_alunos
Nonparametric Statistics and Machine Learning Methods	841	4	24

```
dbDisconnect(conn)
```

```

Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()

```

```
[1] "2026-09-23 16:53:50 -03"
```